Stream processing on yelp dataset

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Agenda

1. Introduction
2. Yelp Data set
3. Infrastructure for the stream processing
4. Stream processing and results on the data
    a. Sentiment analysis on reviews data
    b. Check in and busiest day analysis for business
    c. Famous business categories with respect to location
    d. Covid-19 effects on the business
5. Demo

Introduction

Use case:

The requirements of customers vary day to day. Businesses need to learn about their target market to make the efficient use of resources.

Examples:

1. A restaurant can utilize the data analytics tools to check their peak hours for their business
2. A grocery store can learn about the trend of their customers to bring the best services or products in the market

Benefits of business analytics and intelligence

1. Better decision making
2. Improved operational efficiency
3. Increased competitive advantage
4. Growth in revenues
5. Enhanced customer service
6. Compliance and risk management



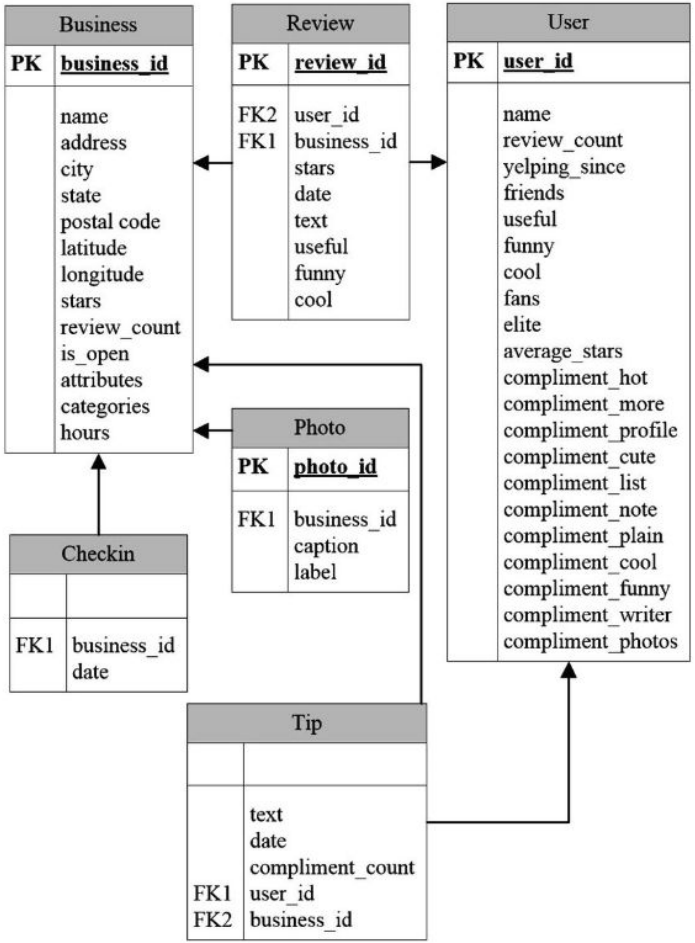
Yelp data set

Yelp data set has the following files.

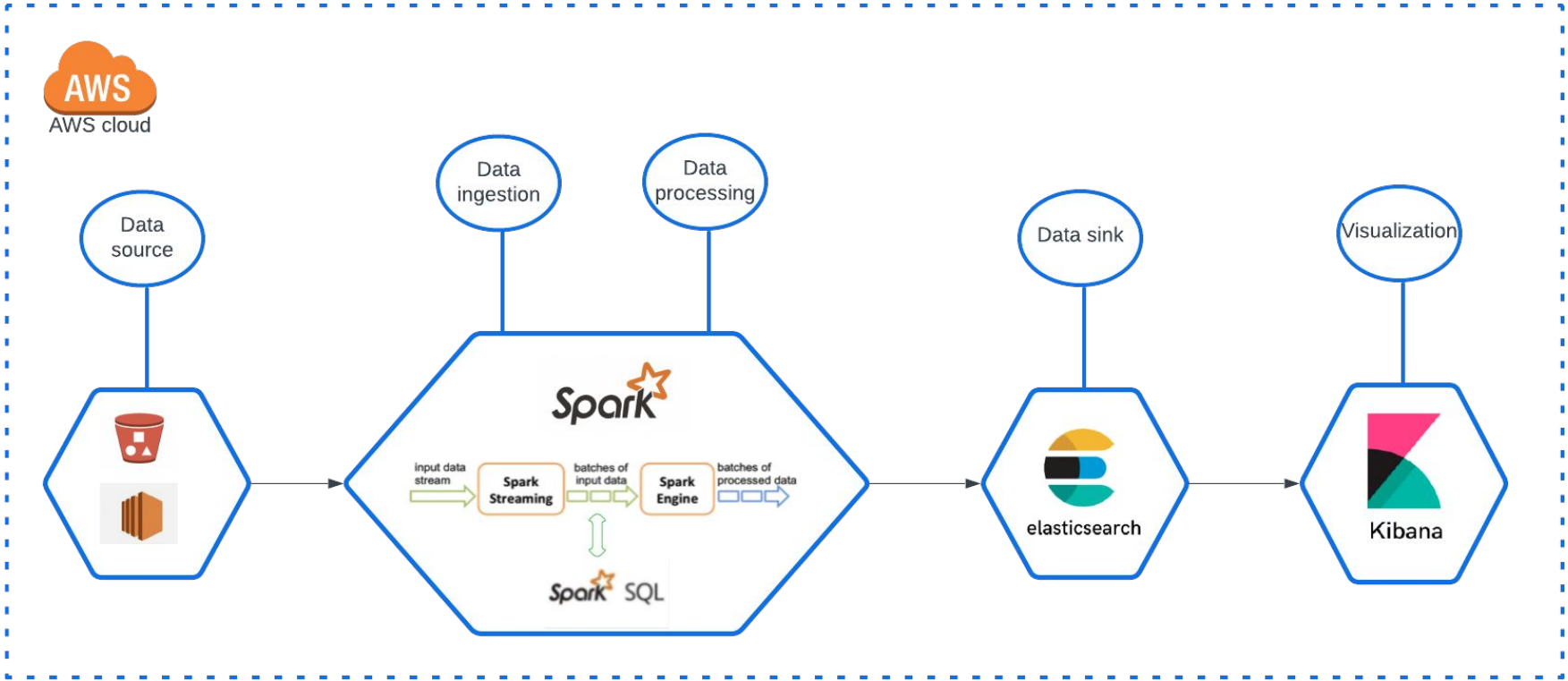
1. Business.json
2. Review.json
3. Check-in.json
4. Tips.json
5. Users.json

Note: For better efficiency of the System,

I converted Review and check-in files to CSV



Architecture for the stream analysis on the yelp data set



Resources Used from AWS

1. 1 Instance for Spark Master node
2. 2 Instances for Spark Worker nodes
3. 1 Instance for Elasticsearch and kibana
4. S3 bucket

Business metrics calculations using structured streaming

Main focus of the analytics:

1. Find the busiest days for any business
2. Weighted sentiment analysis of the business based on user reviews
3. Effect on business activity during Covid-19 peak years
4. Opportunities for new businesses based on the location

Busiest day analysis

Streaming data source: User Check-ins

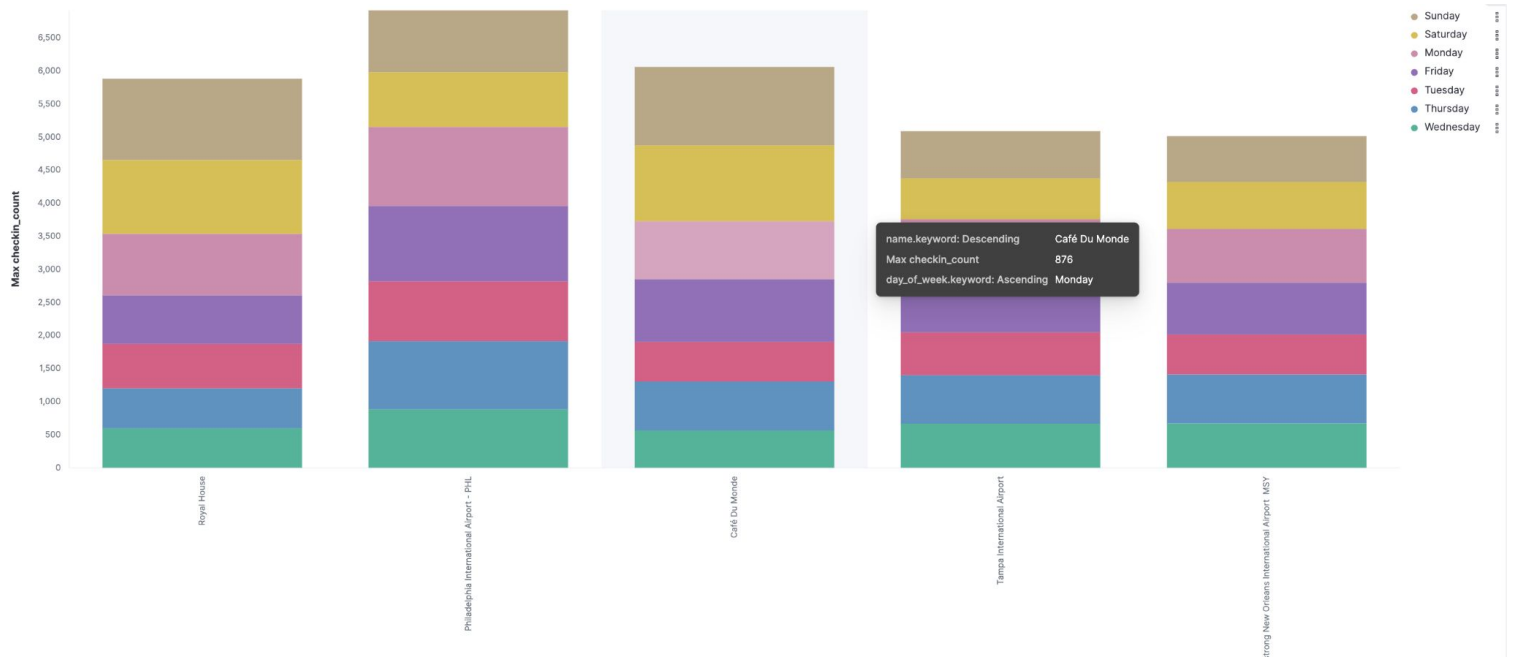
Static data source: Business data

Analytics technique: Temporal data analytics based on year month and day

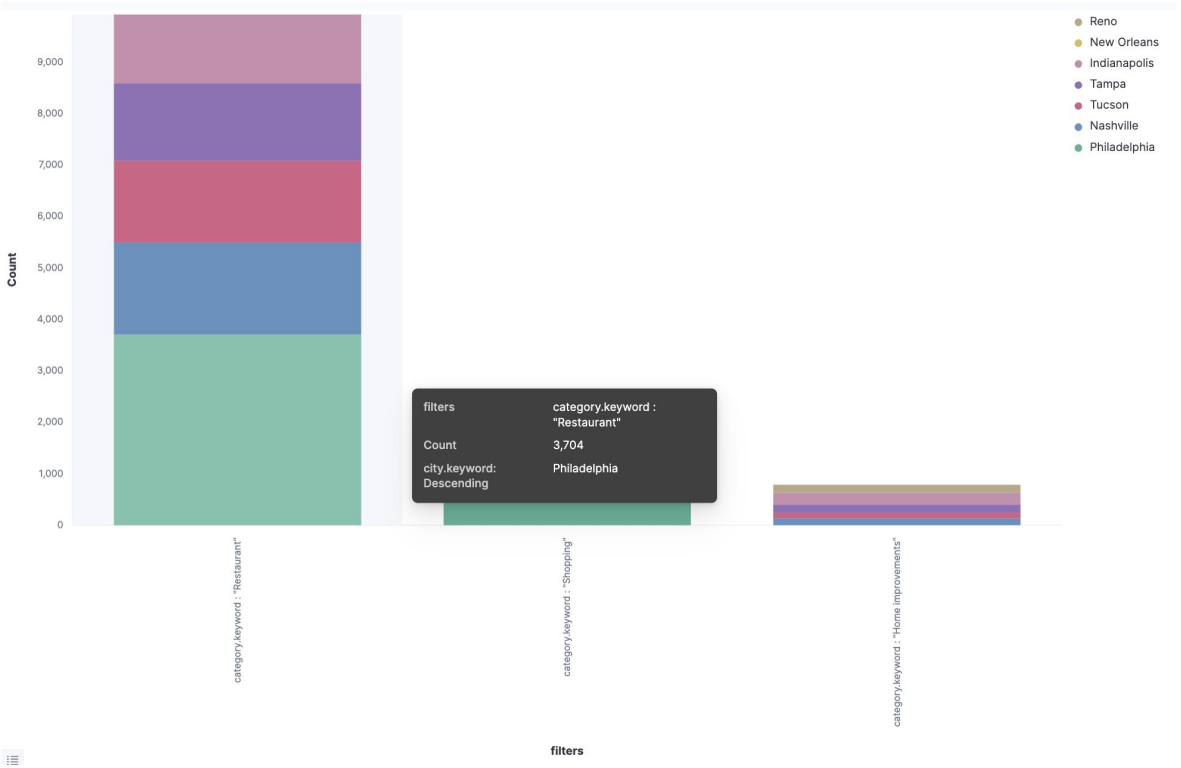
```
def processBatch(batch_df, batch_id, business_dstream):
    batch_df = batch_df.withColumn('year', year("date"))
                       .withColumn('month', month("date"))
                       .withColumn('checkin_time', col('date'))
                       .withColumn('day_of_week', dayofweek("date"))
                       .withColumn('day_of_week', date_format('date', 'EEEE'))
                       .withColumn('event_time', date_trunc('day', col('date')).cast(TimestampType()))

    batch_df = batch_df.withWatermark('event_time', '365 days') \
        .repartition('business_id') \
        .groupBy(window('event_time', '365 days', '365 days'), 'business_id', 'day_of_week') \
        .agg(count('*').alias('checkin_count'),
            first('year').alias('year'),
            first('month').alias('month'),
            first('checkin_time').alias('checkin_time'))
```

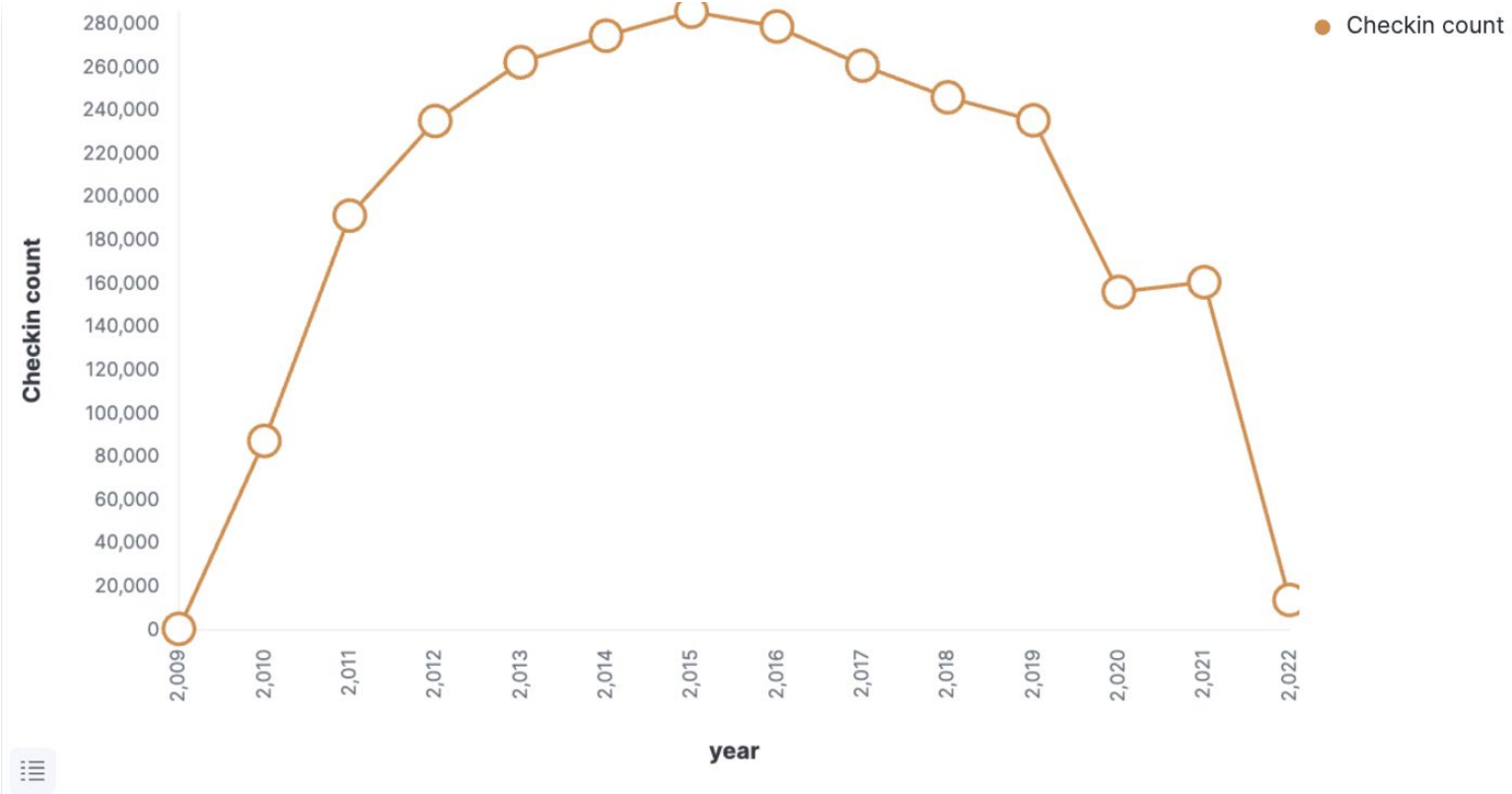
Busiest days for top 5 businesses



Opportunities for new business based on location



Covid-19 effect on business check-ins



Weighted sentiment analysis

Streaming data: User reviews data

Static data source: Business data

Analytics technique:

1. Sentiment value calculation against each review
2. Temporal analysis
3. Weighted sentiment analysis using decay factor

Formulas for weighted sentiment analysis

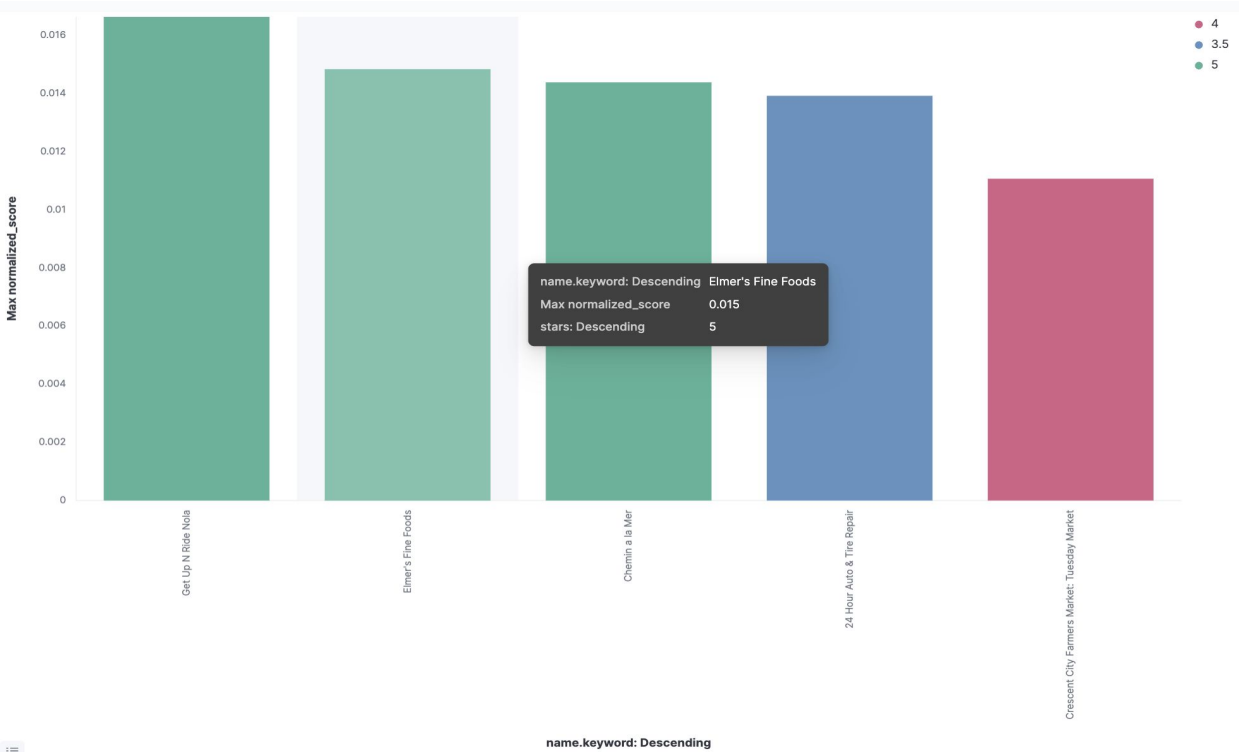
1. Assigned weighted value to each review based on the current timestamp difference with the review time.
2. Calculated weighted score for each sentiment based on decay factor, usefulness of the review and the sentiment score calculated using natural language processing tool
3. Normalized score based on total reviews for the business

$$decay_factor = decay_rate \times DATEDIFF(current_time_stamp, date)$$

$$weighted_score = sentiment_score \times (useful + 0.1) \times decay_factor$$

$$normalized_score = weighted_score / review_count$$

Normalized sentiment score vs star rating for top 5 business



Demo

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Questions